Gravitation.

→ **Gravitation** - Force of attraction between any two masses.

- It is one of the fundamental forces of nature.

→ **Gravity** - Force of attraction between earth and any other mass.

→ **Universal Law of Gravitation/Newton's Law of Gravitation**

- $F \propto m_1$

$F \propto m_2$

$F \propto \frac{1}{r^2}$ → Inverse squar Law.

$F \propto \frac{m_1 m_2}{r^2}$

$F = \frac{G m_1 m_2}{r^2}$ → (Universal Gravitational. Constant).

→ According to the law of gravitation the force of attraction between any two masseso is

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directly proportional to the product of two masses and inversely proportional to the square of the distance between them.

⇒ Now find (G)

If $m_1 = m_2 = m = 1 kg$

and $r = 1 m$

$\boxed{F = G}$

⇒ Unit of G

⇒ $F = \frac{G m_1 m_2}{r^2}$

⇒ $G = \frac{F r^2}{m_1 m_2} = Nm^2/kg^2$

☆⇒ Unit of (G) = $\boxed{Nm^2/kg^2}$

☆⇒ Value of (G) = $\boxed{6.67 \times 10^{-11} Nm^2/kg^2}$

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Q) If the Force of the attraction between two masses seperated by the distance of r is f.

1) If $r = \frac{r}{2}$, What will be the force.

2) If $r = 2r^2$, What will be the force.

Sol ① $F = \frac{Gm_1m_2}{r^2}$ — ①

- Now, When $r = \frac{r}{2}$ (halved).

$$F' = \frac{Gm_1m_2}{\left(\frac{r}{2}\right)^2} = \frac{Gm_1m_2}{\frac{r^2}{4}}$$

$$F' = \frac{4Gm_1m_2}{r^2} = 4F$$

Sol ② $r = 2r^2$

$F = \frac{Gm_1m_2}{2r^2}$ — ①

- Now when $r = 2r$

$$F' = \frac{Gm_1m_2}{(2r)^2} = \frac{F}{4}$$

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⇒ Features of Gravitational Force/characteristics:-

1) It is always attractive in nature.

2) It is long range force but weakest in nature.

3) It obeys inverse square law. $\left[\frac{1}{r^2}\right]$

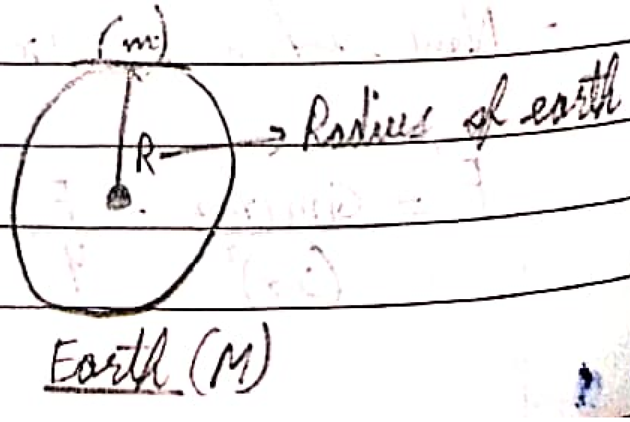
4) It is a central force i.e; It exists along the line joining the centre of the two bodies.

⇒ Acceleration due to Gravity -

- Acceleration is because of Gravitational pull.

⇒ Free Fall - If the object falls from certain height under the action of gravity alone is called free fall.

⇒ Expression for (g) -



$\therefore \quad F = \frac{GMm}{R^2}$ — (I) $\left[\begin{array}{l} M = \text{Mass of earth} \\ m = \text{Mass of object} \end{array}\right]$

- Now Acc. to Newtons law of Motion.

$$F = mg \quad \text{— (II)}$$

⇒ Compare (I) and (II)

$$\cancel{m}g = \frac{GM\cancel{m}}{R^2} \Rightarrow \boxed{g = \frac{GM}{R^2}}$$ → Acceleration due to gravity on surface of earth.

→ Now value of (g) on the surface of earth.

$$\Rightarrow g = \frac{GM}{R^2} = \frac{6.67\times10^{-11}\times6\times10^{24}\,kg}{(6.4\times10^{6})^2}$$

$$= \frac{40.02\times10^{13}}{40.96\times10^{12}} = 0.981\times10^{13}\times10^{-12}$$

$$= 0.981\times10^{1}$$

$$\boxed{g = 9.8\,m/s^2}$$

$$\Rightarrow \boxed{g = \frac{GM}{R^2}}$$

⇒ ∴ g depends on Mass of Planet not on mass of body.

→ **Variation of (g) with shape of Earth:-**

As we know :- $g = \frac{GM}{R^2}$

$$R_{eq} > R_{pol} \qquad \boxed{g \propto \frac{1}{R}}$$
$$g_{eq} < g_{pol}$$

∴ If we move an object from equator to pole what will happen to its weight.

$W = mg$ ∴ Weight will increase.

⇒ **Value of (g) on the surface of moon:-**

→ $g = \frac{GM}{R^2}$

$G = 6.67 \times 10^{-11}$
$M = 7.3 \times 10^{22}$ kg
$R = 1.74 \times 10^{6}$

⇒ $g = \frac{6.67 \times 10^{-11} \times 7.3 \times 10^{22}}{(1.74 \times 10^{6})}$

$g = \frac{48.68 \times 10^{11}}{3.02 \times 10^{12}}$

$= 16.11 \times 10^{11} \times 10^{-12}$

$= 16.11 \times 10^{-1}$

$$\boxed{g_m = 1.6 \, m/s^2}$$

$\therefore \dfrac{g_e}{g_m} = \dfrac{9.8}{1.6} = \dfrac{6}{1} = 6$

$\boxed{g_e = 6 g_m} \rightarrow \boxed{g_m = \dfrac{1}{6^{th}} g_e}$

⇒ Motion of the object under (g):

→ $v = u + at$	Under gravity. $(a = g)$ $v = u - gt$
→ $s = ut + \frac{1}{2}at^2$	$h = ut - \frac{1}{2}gt^2$
→ $v^2 - u^2 = 2as$	$v^2 - u^2 = 2gs$

Q) An object is thrown vertically upwards and rised to a height of 10m. Calculate the velocity with which the object was thrown upwards and the time taken by object to reach the highest point.

Sol) $v^2 - u^2 = -2gh$

$v = 0$

$+u^2 = +2gh$

$u^2 = 2gh$

$u = \sqrt{2gh}$ $\quad (g = 9.8\, m/s)$

$u = \sqrt{2 \times 9.8 \times 10}$

$= \sqrt{196}$

$= 14\, m/s$

⇒ Time:

$\cancel{v = u - gt}$ $\quad v = u - gt$

$\cancel{t = 14/g}$ $\quad 0 = 14 - 9.8 \times t$

$9.8t = 14$

$t = \frac{14}{9.8} = 1.4s$

Q15) A stone is thrown vertically upwards with a initial velocity of 40 m/s, taking $g = 10 m/s^2$. Find the maximum height reached by the stone

What is the net displacement and the total distance covered.

Sol.) $u = 40 m/s$
$v = 0$
$g = 10 m/s^2$

$\Rightarrow$ $v^2 - u^2 = -2gh$
at maximus height $v = 0$

$\Rightarrow$ $\cancel{-}u^2 = \cancel{-}2 \times 10 \times h$
$40^2 = 20h$
$1600 = 20h$
$h = \frac{1600}{20}$

$h = 80 m.$

$\Rightarrow$ Net displacement $= 0$
Distance $= 160 m.$

Q14) A stone is released from the top of a tower of height $= 19.6 m$. Calculate its final velocity

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just before touching the ground.

Sol) $u = 0$

$h = -19.6$

$g = 9.8$

$\Rightarrow -u^2 - v^2 = -2gh$

$v^2 = -2gh$

$v^2 = +2 \times 9.8 \times 19.6$

$v^2 = 384.16$

$v = \sqrt{384.16}$ m/s.

Q16: Calculate the force of gravitation between the earth and the sun, Given that the mass of earth $= 6 \times 10^{24}$ kg. and mass of sun $= 2 \times 10^{3}$ kg. The average distance between the two is 15×10^{11} m.

Sol: Mass of earth $= 6 \times 10^{24}$

Mass of sun $= \cancel{6 \times 10^{24}}\ 2 \times 10^{3}$

Distance $= 1.5 \times 10^{11}$ m

$\Rightarrow F = \frac{GMm}{r^2}$

$$F = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 2 \times 10^{13}}{(1.5 \times 10^{11})^2} = \frac{80.05 \times 10^{43} \times 10^{-22}}{2.25}$$

$$\boxed{F = 35.57 \times 10^{21} N}$$

Q17) A stone is ~~about~~ allowed to fall from the top of the tower 100m height and at the same time, another stone is projected vertically upwards from the ground with a velocity of 25 m/s. Calculate when and where the two stones will meet.

Sol) $u = 0$

$SA = ut + \frac{1}{2}gt^2$

$SA = \frac{1}{2}gt^2$ — (I)

$\Rightarrow SB = ut - \frac{1}{2}gt^2$

$u = 25 m/s$

$\Rightarrow SB = 25t - \frac{1}{2}gt^2$ — (II)

$\Rightarrow SA + SB = 100$

$\cancel{\frac{1}{2}gt^2} + 25t - \cancel{\frac{1}{2}gt^2} = 100$

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$25t = 100$

$$\boxed{t = \frac{100}{25} = 4s}$$

Now, $S_A = \frac{1}{2} g t^2$

$$S_A = \frac{1}{\cancel{2}_1} \times \cancel{10}^5 \times 4^2$$

$$= 5 \times 16 = 80\,m.$$

$\therefore$ Distance from ground where A and B meets

$$\Rightarrow \boxed{100 - 80 = 20\,m}$$